

Joydeep Das

+91-7085917058 | joy.thesloth@gmail.com
github.com/JoyTheSloth | [LinkedIn](#) | [Portfolio](#)



PROFESSIONAL SUMMARY

GenAI & LLM Engineer specializing in RAG systems, multi-agent AI, and full-stack AI applications. Built MediRAG — a hallucination-resistant clinical AI system grounded in 1.1M+ medical records — and multiple production-grade AI systems spanning autonomous agents, health intelligence, and real-time research pipelines. Experienced with LangChain, LangServe, Ollama, Groq, and Google Gemini API. Passionate about building real-world AI products that scale.

EDUCATION

Amity University Kolkata, India | *July 2022 – Present*

- Bachelor of Technology – Computer Science and Engineering; GPA: 7.1/10

Unacco School Imphal West | *2022*

- Class XII, Percentage: 82%

Heritage Convent Imphal West | *2020*

- Class X, Percentage: 70%

TECHNICAL SKILLS

- **AI / LLM:** LangChain, RAG, Prompt Engineering, Multi-Agent Systems, LangServe, LangSmith, Ollama (Gemma, LLaMA 3), Groq, Hugging Face, Vector Databases, Generative AI
- **Backend:** Python, FastAPI, Node.js, Express.js, Flask, RESTful APIs
- **Frontend:** React.js, Next.js, Tailwind CSS, JavaScript (ES6+), HTML5, CSS3
- **ML / Data:** TensorFlow, Keras, Scikit-learn, Pandas, NumPy, BioBERT, FAISS, BM25
- **Tools & Infra:** Git, GitHub, Supabase, Vercel, Hugging Face Spaces, Make.com, n8n

PROJECTS

MediRAG: Clinical-Grade AI Reasoning & Retrieval | *India Innovates 2026 · Rank 62nd nationally · 2026*

- Built a clinical AI system on 1.1M+ medical records with 107k+ indexed clinical vectors, reducing hallucinations via a 2-step dual-LLM verification pipeline with NLI-based faithfulness checks
- Hybrid retrieval engine combining BioBERT (semantic understanding) and BM25 (keyword matching) for high-precision medical Q&A
- Document-aware MediChat for PDF/report analysis with rich source citations and relevance scores

FitnessBunny: CalPal Freemium | *Next.js 16, Supabase, Groq · 2026*

- Replaced manual calorie logging with automated nutritional analysis using Groq-powered LLMs, improving tracking efficiency by ~60%
- AI-powered recipe generation engine transforming raw ingredients into personalized meal plans
- Full-stack architecture with Next.js 16, Supabase (PostgreSQL), and NextAuth using a database-first RAG approach

Researchix: Autonomous Research Agent System | *Groq, Tavily, arXiv, React · 2026*

- 4-phase agentic pipeline (plan → retrieve → analyze → synthesize) processing academic content autonomously
- Real-time academic retrieval via Tavily Search and arXiv APIs with intelligent source filtering
- Full-stack: Express/Node.js backend on Hugging Face Spaces, React/Vite/Tailwind frontend on Vercel

Multi-Agent Bug Analysis System (MABAS) | *Groq + FastAPI + React · 2026*

- 5-agent autonomous pipeline (triage → reproduce → analyze → patch → review) resolving bugs end-to-end with risk scoring (0–100)
- Dynamic bug reproduction via sandboxed Python execution generating real crash scenarios

Multi-Agent Launch Decision System (MLDS) | *2026*

- Autonomous 6-agent system (Data, Marketing, Product, Engineering, Risk, Coordinator) generating real-time Proceed/Pause rollout decisions
- Live React dashboard visualizing agent reasoning traces and tool outputs in real-time

Bacterial Species Classification System | *2025*

- Bacterial classification model identifying 5 species with 95.83% test accuracy using VGG16 + SVM
- PCA-based dimensionality reduction; CLI tool and Python API for single-image predictions

Additional projects (Vibe Check AI Weather App, Trending Movies Platform, Library Management API) available at github.com/JoyTheSloth

ACHIEVEMENTS & COMPETITIONS

India Innovates 2026 – National Hackathon | *Open Innovation Category · Rank 62nd nationally · 2026*

- Built MediRAG — an AI healthcare system on 1.1M+ curated medical records — achieving national recognition
- Deployed backend on Hugging Face Spaces; integrated FastAPI REST APIs for real-time inference

WORK EXPERIENCE

Lead UI/UX Designer | *GEETBIH Labs Pvt. Ltd. · Part-time · April 2026 – Present*

- Driving end-to-end design for digital products with a strong focus on usability, scalability, and impactful user experiences
- Leading creation of intuitive interfaces while building and maintaining a cohesive design system across web and mobile platforms
- Leveraging generative AI and prompt design to streamline workflows and accelerate the design process

App Designer | *Modern Mahal · August 2025 – November 2025*

- Designed intuitive, visually appealing web and mobile interfaces in Figma to align with brand identity
- Conducted usability reviews and implemented design refinements to improve navigation and accessibility

Generative AI Developer Intern | *AI Wallah · Remote · May 2025 – July 2025*

- Developed and deployed LLM-powered web applications using Google Gemini API for Q&A and multi-turn conversational workflows
- Built full-stack integration of LLM services with Flask backend and JavaScript frontend via RESTful APIs
- Accelerated AI feature development by ~50% through optimized prompt engineering and rapid prototyping

UI/UX Designer & Social Media Marketing | *2GatherApp · Remote · February 2025 – October 2025*

- Designed user-centric interfaces that enhance community engagement and streamline event discovery
- Built a consistent design system to unify app visuals and marketing assets
- Created and executed social media campaigns to boost reach, downloads, and brand identity

Web Designer | *YGSD · September 2024 – November 2024*

- Project: Mourya Urja Matrimonial — Designed user-friendly web layouts in Figma for a trusted matrimonial brand; integrated client feedback to enhance UX
- Project: Veliciae Jewellery — Created visually cohesive web designs in Figma; collaborated with developers on effective design solutions

CERTIFICATIONS

Career Essentials in GitHub Professional Certificate | *GitHub & LinkedIn Learning · 2025*

Build Your Generative AI Productivity Skills | *Microsoft & LinkedIn Learning · 2024*

Career Essentials in Generative AI | *Microsoft & LinkedIn Learning · 2024*